

Question Bank

Unit III: Principles of Numerical Control and Coding (12 Hrs.)

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Syllabus: Basic principles of automation; extending the capabilities of conventional machines through improved devices and manipulators; basic principles of numerical control; CNC, DNC and machining centres; methods of coding and programming; APT programming; adaptive control; economics of numerical control.

Part A 10 multiple-choice questions $10 \times 1 = 10$ marks

Part B 5 short-answer questions $5 \times 2 = 10$ marks

Part C 3 long-answer questions $3 \times 10 = 30$ marks

Part A: Multiple-Choice Questions

1 mark each

Q1. In an NC machine tool, the Z axis is always

- (a) the longest horizontal axis
- (b) a rotary axis
- (c) vertical
- (d) along the spindle axis

Answer: (d) Along the spindle axis; +Z moves the tool away from the workpiece.

Q2. The G-code G01 commands

- (a) rapid positioning
- (b) linear interpolation at feed rate
- (c) circular interpolation CW
- (d) dwell

Answer: (b) Linear interpolation (straight-line cutting move) at the programmed feed.

Q3. The M-code M03 means

- (a) spindle ON, clockwise
- (b) program stop
- (c) coolant ON
- (d) tool change

Answer: (a) Spindle ON, clockwise.

Q4. Drilling a pattern of holes needs which type of NC control?

- (a) Adaptive
- (b) Continuous path
- (c) Point-to-point
- (d) Manual

Answer: (c) Point-to-point — the path between holes does not matter.

Q5. APT stands for

- (a) Automatic Part Tooling
- (b) Automatically Programmed Tools
- (c) Advanced Programming Technique
- (d) Absolute Position Transfer

Answer: (b) Automatically Programmed Tools.

Q6. In DNC, a central computer

- (a) replaces the machine tool
- (b) is used only for drawing
- (c) controls only one spindle
- (d) stores programs and sends them to many machines

Answer: (d) Stores part programs and distributes them to many CNC machines over a network.

Q7. The device that changes tools automatically on a machining centre is called

- (a) ATC (b) APC
(c) DRO (d) MCU

Answer: (a) ATC — Automatic Tool Changer.

Q8. The G-code for absolute programming is

- (a) G17 (b) G91
(c) G90 (d) G28

Answer: (c) G90 (G91 is incremental).

Q9. Adaptive control constraint (ACC) systems keep

- (a) a process variable such as cutting force at or below a limit (b) the spindle speed constant always
(c) the program unchanged (d) the tool cold

Answer: (a) A measured variable (force, power, torque) at or below a set limit by adjusting feed/speed.

Q10. The first NC milling machine (1952) was developed at

- (a) IIT Bombay (b) MIT
(c) Cambridge (d) Tokyo University

Answer: (b) MIT (with J. T. Parsons, for the US Air Force).

Part B: Short-Answer Questions

2 marks each

Q1. Define numerical control and name its three basic components.

[2]

Answer

Numerical control (NC) is a form of programmable automation in which the motions of a machine tool are controlled by a program of coded numbers, letters and symbols. Its three basic components are: (1) the **part program**, (2) the **machine control unit (MCU)**, and (3) the **machine tool** (processing equipment).

Q2. Distinguish between absolute and incremental positioning with an example.

[2]

Answer

In **absolute** positioning (G90) every coordinate is measured from the program zero; in **incremental** positioning (G91) each move is measured from the previous position.
Example: moving from A(10, 10) to B(40, 10): absolute X40 Y10; incremental X30 Y0.

Q3. An open-loop NC axis uses a stepper motor of 200 steps/rev directly coupled to a lead screw of 5 mm pitch. Find the BLU and the pulses needed for a 100 mm move.

[2]

Answer

$$\text{BLU} = \frac{p}{n_s} = \frac{5}{200} = 0.025 \text{ mm.} \quad \text{Pulses} = \frac{100}{0.025} = 4000.$$

Q4. Explain the word-address block format with an example.

[2]

Answer

In the word-address format every word begins with a **letter address** that tells the controller the meaning of the number that follows; words may be in any order and only changed words need be written.

Example: N050 G01 X25.0 Y30.0 F200 S1000 M03 — block 50, linear feed to (25, 30) at 200 mm/min, spindle 1000 rpm clockwise.

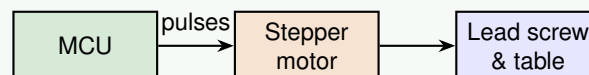
- Q5.** A CNC machine costs Rs 40 lakh and gives net savings of Rs 10 lakh per year. Find the payback period and state its meaning. [2]

Answer

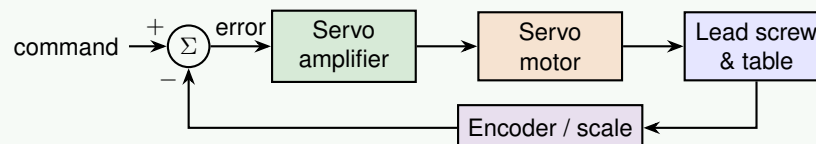
Payback = $\frac{\text{investment}}{\text{net annual saving}} = \frac{40}{10} = 4 \text{ years}$. It is the time needed for the savings to recover the initial investment; a shorter payback means a more attractive investment.

Part C: Long-Answer Questions**10 marks each**

- Q1.** With neat block diagrams, explain open-loop and closed-loop NC systems. Explain the basic components of a CNC system and list the advantages of CNC over conventional NC. [10]

Answer**(a) Open-loop NC system**

The MCU sends a counted number of pulses to a stepper motor; each pulse moves the table one BLU. There is **no feedback**, so missed steps (overload) are not detected. It is simple and cheap and is used for light-duty machines.

(b) Closed-loop NC system

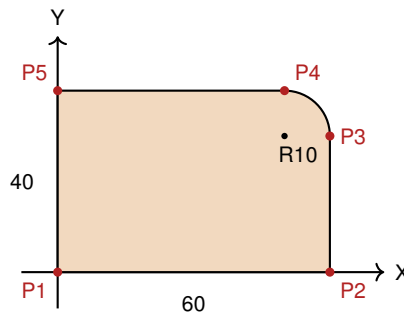
A servo motor drives the table; an **encoder or linear scale** measures the actual position and feeds it back. The MCU compares it with the command and corrects the error, giving high accuracy even under varying loads.

(c) Components of a CNC system

- **Program input** — keyboard (MDI), USB, network/DNC.
- **CPU and memory** — store the program, decode blocks, perform **interpolation** and compensation.
- **Servo drives** — one per axis plus the spindle drive.
- **PLC section** — handles M-codes: spindle, coolant, tool changer, door interlocks.
- **Feedback devices** — encoders, tachometers, linear scales.
- **Operator panel / display** — graphics, diagnostics.

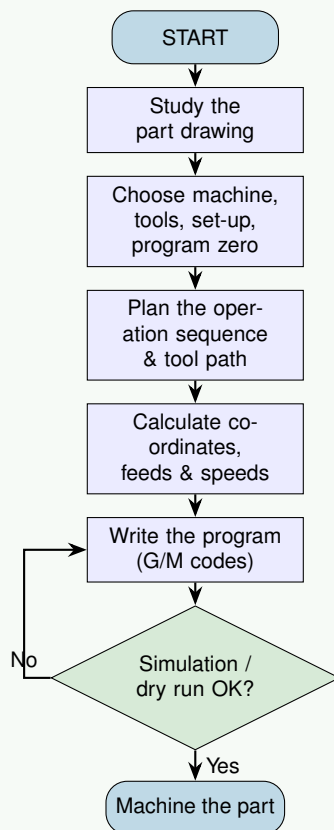
(d) Advantages of CNC over hard-wired NC: program stored and edited at the machine; any interpolation in software; tool-length and cutter-radius compensation; canned cycles and sub-programs; graphic simulation; diagnostics; easy DNC networking; one program runs many parts without re-reading tape.

- Q2.** Explain the steps of manual part programming with a flowchart. Write a word-address (G-code) part program to mill the outer profile shown, using a 10 mm end mill, depth 2 mm, spindle 1200 rpm and feed 200 mm/min. [10]



Answer

(a) Steps in manual part programming



(b) Part program (program zero at P1, top face Z0; tool centre on the outline for simplicity)

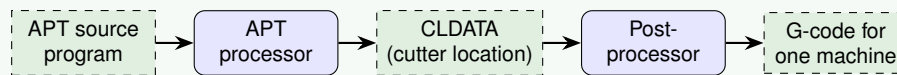
Block	Meaning
O1001	program number
N10 G21 G90 G17	metric, absolute, XY plane
N20 T01 M06	load tool 1 (10 mm end mill)
N30 G00 X0 Y0 Z5.0	rapid above P1
N40 S1200 M03 M08	spindle CW, coolant ON
N50 G01 Z-2.0 F100	plunge 2 mm
N60 X60.0 F200	P1 → P2
N70 Y30.0	P2 → P3
N80 G03 X50.0 Y40.0 R10.0	arc P3 → P4 (CCW)
N90 G01 X0	P4 → P5
N100 Y0	P5 → P1
N110 G00 Z50.0 M09	retract, coolant OFF
N120 M05	spindle stop
N130 M30	end, rewind

Notes: G01/G00 and X/Y words are **modal** — they stay active until changed, so repeated codes are omitted. The arc from P3 to P4 is counter-clockwise about the centre (50, 30), hence G03. To cut the true outline with the tool outside the part, cutter-radius compensation G41/G42 with an offset of 5 mm would be added.

- Q3.** Explain APT programming: the structure of an APT system (with a block diagram), the four types of APT statements, and the motion statements. Write the APT motion statements for contouring the part of Q2. [10]

Answer

(a) APT system



APT (Automatically Programmed Tools, MIT 1956–59) lets the programmer describe the **geometry** and **tool motion** in English-like statements. The **processor** computes the cutter-centre path (CLDATA), independent of the machine. The **post-processor** converts CLDATA into the codes of a particular machine/controller.

(b) Four types of statements

Type	Purpose	Example
Geometry	define points, lines, circles, planes	P1 = POINT/0, 0, 0; L1 = LINE/P1, P2
Motion	describe the tool path	FROM/P0; GORGT/L1, PAST, L2
Post-processor	speeds, feeds, coolant	SPINDL/1200, CLW; FEDRAT/200
Auxiliary	part name, cutter, tolerance, end	PARTNO; CUTTER/10; FINI

(c) Motion statements: point-to-point FROM/, GOTO/, GODLTA/; contouring direction words GOLFT, GORGT, GOFWD, GOBACK, GOUP, GODOWN; check-surface modifiers TO, ON, PAST, TANTO; tool side TLLFT, TLRGT, TLON.

(d) APT program for the part of Q2 (L1: $y = 0$, L2: $x = 60$, C1: centre (50, 30) R10, L3: $y = 40$, L4: $x = 0$, PL1: $z = -2$, start point P0 (−20, −20, 10))

Statement	Meaning
CUTTER/10.0	10 mm cutter
SPINDL/1200, CLW FEDRAT/200	spindle and feed
FROM/P0	start point
GO/TO, L1, TO, PL1, TO, L4	go down to touch L1 and L4 at depth PL1
TLRGT, GORGT/L1, PAST, L2	tool right of L1, cut to beyond L2
GOLFT/L2, TANTO, C1	turn left along L2 up to tangent with C1
GOFWD/C1, TANTO, L3	follow the arc to tangent with L3
GOFWD/L3, PAST, L4	along L3 to beyond L4
GOLFT/L4, PAST, L1	down L4 to beyond L1
GOTO/P0 FINI	return, end